

Akshat Sureshbhai Desai

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EDUCATION

California State University, Fullerton | Fullerton, CA

Aug 2024 - May 2026

Master of Science in Computer Science | GPA: 3.65/4.0

Charotar University of Science and Technology | Ahmedabad, India

Nov 2020 - May 2024

Bachelor of Technology in Computer Science | GPA: 3.6/4.0

SKILLS

Languages: Python, C++, SQL, Bash

ML/AI: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, ResNet, CNNs, Computer Vision, Medical Imaging, Time-Series Forecasting, Feature Engineering

LLM/GenAI: vLLM, LoRA, Fine-tuning, RAG, LangChain, LangGraph, ChromaDB, FAISS, Hugging Face, LLM Eval

MLOps/Cloud: Docker, Kubernetes, GCP Cloud Run, AWS Bedrock, AWS EC2, Firebase, Quantization, Inference Optimization, Flask, FastAPI

Systems/Data: Linux, HPC/SLURM, Git/GitHub, REST APIs, Pandas, NumPy, PyRadiomics, FreeSurfer

PROFESSIONAL EXPERIENCE

AI/LLM Research Developer | CSUF | Fullerton, CA

June 2025 - May 2026

- Deployed a production **Verilog AI tutor** used by CSUF students and faculty for hardware design generation and simulation.
- Eliminated timing errors by fine-tuning a 32B code LLM on **1,000 compiler-verified** examples using **LoRA** on H200 GPU.
- Cut student onboarding **time by ~45 minutes** for **100+ users** by shipping a browser-based simulator with Firebase auth, compile logs, and live waveform previews.
- Compressed model footprint by **70% by quantizing** the fine-tuned model from **64GB to 19.22GB** for efficient vLLM inference.
- Reduced generation failures by implementing dual-corpus RAG - **14,740 compiler-verified** Verilog examples in ChromaDB.

Lead AI Developer | Worthwhile LLC | Escondido, CA

Oct 2025 - Jan 2026

- Enabled **real-time voice interaction** for Lightwall, a museum AI art installation, by building an offline **Llama/Whisper/radar pipeline** with **sub-second latency**.
- Turned **visitor proximity** and **LLM mood** into live physical behavior, controlling LED colors, stepper motors, and personality states through hardware-control tags.
- Grounded responses and **reduced hallucinations** by injecting project memory and live codebase context into system prompts.

Machine Learning Research Associate | CSUF | Fullerton, CA

Nov 2024 - May 2025

- Led Alzheimer's MRI radiomics research to classify CN, MCI, and AD patients from **3D brain scans** using machine learning.
- Increased usable patient cohort from **275 to 1,229 ADNI subjects (4.5x)** by building a custom pipeline that reconstructed fragmented MRI subfield outputs into complete patient-level feature vectors.
- Cut MRI **processing time by 50%** by automating segmentation and feature extraction on HPC clusters using SLURM array jobs.
- Achieved **92.73%** accuracy by reducing **5,980 radiomics features to 77 biomarkers** via LASSO and training **XGBoost**.

Computer Vision & Imaging Intern | Space Applications Centre, ISRO | Ahmedabad, India

Dec 2023 - May 2024

- Built a real-time satellite-camera exposure-control pipeline at ISRO, achieving **near-accurate correction at 2 fps** on live camera streams with **no pre-existing** dataset or deep learning required.
- Evaluated DSOD and DNN-based exposure methods, then developed a **dataset-free skewness** algorithm using histogram third-moment analysis.

PROJECT EXPERIENCE

CPP Dining AI Staffing Dashboard | AWS Hackathon | Flask, XGBoost, AWS Bedrock, LangChain, LangGraph

- Won **1st place** at the Cal Poly Pomona AWS Hackathon by building an AI staffing dashboard for campus dining operations.
- Forecasted **six-role** dining staffing with XGBoost using operational features, reducing projected annual labor waste by **\$52,000**.
- Built an **AWS Bedrock** chatbot using **LangChain** and **LangGraph** via Flask **APIs** to explain predictions and surface staffing recommendations in plain language.

Multi-Class Breast Cancer Subtype Classification | Python, PyTorch, ResNet, Grad-CAM, Medical Imaging

- Built an AI diagnostic pipeline to classify 8 breast cancer subtypes from **7,909 biopsy** images using **PyTorch** transfer learning.
- Outperformed prior breast cancer subtype classification methods by **~5% accuracy** and **~4% AUC-ROC**, achieving **91.54%** accuracy, 99.68% AUC-ROC, and 98.67% specificity with ResNet-50.
- Applied **Grad-CAM** to highlight tumor-relevant regions for clinical interpretability. Published in *MDPI Journal of Imaging*, 2025.

Solar Power Forecasting System | Python, LightGBM, Pandas, NumPy, Time-Series Forecasting

- Built a 15-minute interval **solar power forecasting system** on 6 years of generation and NASA weather data (2019-2025), engineering 39 lag, time, and weather features for short-term energy planning.
- Selected LightGBM over XGBoost, Random Forest, LSTM, and Prophet by **benchmarking** on held-out 2024 test data, achieving **96% variance explained** and **MAE of 1.99 kW** across **210,000+** time-series records.